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# On the Reliability of Microsoft Excel XP for Statistical Purposes

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**Abstract:** The paper deals with the reliability of Microsoft Excel XP for statistical purposes. In section 1 we show that Excel can compute negative variances. In section 2 we see that the random numbers generated by Excel cannot suffice scientific statistical requirements, and in section 3 we have to report that unbearable errors with the computation of statistical distributions that have been reported to Microsoft and later on have been published (Knüsel, 1998) are still to be found in Microsoft Excel XP. So one has again to warn the scientific community against using Microsoft Excel for statistical purposes.

**Keywords:** Microsoft Excel, reliability, accuracy, statistical distributions, random numbers.

## 1. Excel can compute negative variances

Microsoft Excel offers two functions to compute the variance of a series $x_1,\ldots,x_n$ of real numbers: the function *variance* computes the value

of $\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\overline{x})^2$

the function *variancen* computes the value

of $\frac{1}{n}\sum_{i=1}^{n}(x_i-\overline{x})^2$ .

Now for the three numbers $x_1=0, x_2=1, x_3=2$ we have $\overline{x}=1$ and so the correct result with the first formula is 1 and with the second formula $\frac{2}{3}=0.666\ldots67$. The variance does not change, if we add the same constant to all the numbers, i.e. the variance of the three numbers $c, c+1, c+2$ is still $\frac{2}{3}=0.666\ldots67$ (with the second formula) whatever the value of *c*. In Table 1 we give the results of the function *variancen* (in the category of the statistical functions) of Microsoft Excel XP. The results are chaotic, if the constant *c* becomes large, and even negative variances can be obtained!

**Table 1:** Results of *variancen* for the three numbers $c, c+1, c+2$

| *c* | value of *variancen* | comment |
|---|---|---|
| 30E6 | 0.66...67 | result correct |
| 40E6 | 0.88...89 | all 15 digits wrong |
| 160E6 + 1 | −3.555... | negative variance! |
| 1E10 | 14563.555... | all 15 digits wrong |
| 1E10 + 1 | −14563.555... | negative variance |

## 2. Random number generator of Excel questionable

Statistical simulations require good random numbers. Microsoft Excel offers under *Tools, Data Analyses, Random Number Generation* random numbers from the uniform distribution, from the normal distribution, and from some other distributions. Random numbers from the uniform distribution in the interval $[0,1]$ are generated in Excel from integer random numbers from the set $\{0,1,\ldots,32767\}$ by dividing the integer numbers by $m=32767$. One can see this by generating some uniform numbers and by multiplying these numbers with *m*; all the resulting numbers will be integers. So random numbers from the continuous uniform distribution in the interval $[0,1]$ will have only $m+1$ distinct values, namely $0, 1/m, 2/m,\ldots, 1$ if they are generated by Excel. In the view of the author the discrete lattice of these random numbers is not fine enough for scientific statistical puposes.

Random numbers from the standard normal distribution are derived in Excel from random numbers with a uniform distribution; there exists a function that maps a random number from the latter distribution to a random number from the first distribution. So there are again only $m+1$ distinct normal random numbers. The smallest and largest three of these normal random numbers and the corresponding uniform numbers are given in Table 2. To check these facts one can perform the following steps:

1. Generate 10 000 uniform random numbers from $[0,1]$ in each of the 10 columns A-J, using the random seed 31. Compute for each column the maximum and the minimum.
2. Generate 10 000 standard normal random numbers in each of the 10 columns K-T, using again the random seed 31. Compute for each column the maximum and the minimum.

Then one can see that the smallest uniform number corresponds to the smallest normal number and so on.

**Table 2:** The smallest and the largest three random numbers from the uniform and the standard normal distribution

| uniform | normal |
|---|---|
| 0 | −9.5367 4316 |
| $1/m = 0.305\,185 \times 10^{-4}$ | −4.0093 4368 |
| $2/m = 0.610\,370 \times 10^{-4}$ | −3.8417 0562 |
| $\vdots$ | $\vdots$ |
| $(m-2)/m = 0.9999\,3896$ | 3.8417 0562 |
| $(m-1)/m = 0.9999\,6948$ | 4.0093 4368 |
| 1 | 5.3644 1803 |

The function that generates the normal random numbers is obviously not symmetric. And if $Z$ denotes a standard normal random variable, then we have $\Pr\{Z < -9.5367\} = 0.737 \times 10^{-21}$, but with the Excel generator the probability that a standard normal random variable takes on the value −9.5367 is $1/(m+1) = 0.305 \times 10^{-4}$; so this probability is much too large. The findings in this section show that random numbers generated by Excel XP cannot suffice scientific requirements.

## 3. Computation of statistical distributions

In my paper of 1998 (On the accuracy of statistical distributions in Micrsosoft Excel 97) I pointed out a whole list of errors to be found in Excel 97 with the computation of elementary statistical distributions. All these errors are still present in Excel XP with the only exception that the wrong values of the inverse standard normal distribution given by Excel 97 have to be replaced by other values when using Excel XP, but these values are still wrong (see Table 3).

**Table 3:** Inverse standard normal distribution

| x | Value of x such that $\Pr\{X < x\}$ | | |
|---|---|---|---|
| | exact value | with Excel 97 | with Excel XP |
| 0.001 | −3.090 23 | −3.090 24 | −3.090 25 |
| 0.000 1 | −3.719 02 | −3.719 47 | −3.719 09 |
| 1E-5 | −4.264 89 | −4.265 46 | −4.265 04 |
| 1E-6 | −4.753 42 | −4.768 37 | −4.753 67 |
| 3E-7 | −4.991 22 | −7.152 56 (!) | −4.991 52 |
| 2E-7 | −5.068 96 | −5 000 000 (!) | −5.069 28 |

**References:**

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